

HR Analytics

Barplot Insights & Trends

Exploratory analysis of employee workforce data with a focus on distribution, attrition patterns, and categorical comparisons

Dataset	HR Employee Attrition Dataset
Records	1,470 employees
Attributes	35 columns
Target	Attrition (Yes / No)
Analysis	Barplots and categorical comparisons
Tools	Python • Pandas • Matplotlib

Prepared as part of the HR Analytics / People Analytics mentorship project.

1. Executive Summary

The dataset contains 1,470 employee records and 35 attributes. 1,233 employees (83.9%) stayed, while 237 (16.1%) left. The target variable is therefore noticeably imbalanced.

The barplot analysis examines workforce composition and selected categorical variables in relation to attrition. The findings are descriptive: they identify observed patterns and associations, not causal relationships.

Finding	Observed pattern
Attrition	83.9% stayed; 16.1% left.
OverTime	1,054 employees reported no overtime; 416 reported overtime.
Business Travel	Travel_Rarely is the largest group (1,043).
Job Role	Sales Executive is the most represented role (326).

2. Barplot Findings

2.1 Employee Attrition Distribution

Key insight: The dataset is dominated by the 'No' category: 1,233 employees (83.9%) stayed and 237 (16.1%) left.

This imbalance should be considered during predictive modelling because accuracy alone may not adequately describe performance on the minority attrition class.

2.2 Employees by Overtime Status

Key insight: No overtime is the larger workforce category (1,054), compared with 416 employees reporting overtime.

Overtime is useful for subsequent comparison with attrition because its groups can be analysed both by count and by within-group percentage.

2.3 Employee Distribution by Job Role

Key insight: Employee counts vary considerably across roles. Sales Executive has the largest representation with 326 employees.

Unequal group sizes should be considered when comparing attrition across roles.

2.4 Business Travel Distribution

Key insight: Travel_Rarely is the dominant category with 1,043 employees.

Business travel provides a useful workforce segmentation for further attrition analysis.

3. Attrition Comparison

3.1 Job Satisfaction by Attrition

Key insight: Attrition outcomes vary across job-satisfaction levels.

Because satisfaction groups differ in size, within-level percentages are more informative than raw counts alone.

Within-level attrition percentages

Job Satisfaction	Stayed (%)	Left (%)
1	77.2	22.8
2	83.6	16.4
3	83.5	16.5
4	88.7	11.3

3.2 Job Role by Attrition

Key insight: The distribution of attrition outcomes differs across job roles, making job role a useful dimension for further analysis.

Because roles have different workforce sizes, within-role percentages should be considered alongside raw counts.

Within-role attrition percentages

Job Role	Stayed (%)	Left (%)
Healthcare Representative	93.1	6.9
Human Resources	76.9	23.1
Laboratory Technician	76.1	23.9
Manager	95.1	4.9
Manufacturing Director	93.1	6.9
Research Director	97.5	2.5
Research Scientist	83.9	16.1
Sales Executive	82.5	17.5
Sales Representative	60.2	39.8

4. Overtime and Attrition

4.1 Overtime by Attrition

Key insight: Attrition outcomes show a different distribution between employees with and without overtime.

The normalized comparison is more useful than raw counts because the two overtime groups are different in size.

OverTime	Stayed (%)	Left (%)
No	89.6	10.4
Yes	69.5	30.5

The observed relationship supports retaining overtime as a feature for further modelling. It should not be interpreted as evidence that overtime itself causes attrition.

5. Overall Trends & Analytical Interpretation

- **Target imbalance:** 83.9% of employees stayed and 16.1% left.
- **Workforce composition:** Employee counts are uneven across job roles, travel categories, and other segments.
- **Overtime association:** Attrition distributions differ between overtime groups.
- **Job role association:** Attrition percentages vary across job roles.
- **Job satisfaction:** Attrition outcomes vary across satisfaction levels.
- **Descriptive interpretation:** The charts identify patterns and associations; further statistical and predictive analysis is needed for stronger conclusions.

6. Implications for the Next Stage

The identified categorical variables can be carried forward into data preparation and machine-learning workflows. Because the target is imbalanced, later model evaluation should consider precision, recall, F1-score, and the confusion matrix in addition to accuracy.

7. Conclusion

The barplot analysis provides a structured view of workforce composition and selected attrition patterns. Its main value is translating visual patterns into observations that can guide feature selection, modelling, and further HR analytics.

Project note: Findings are based on the supplied HR Employee Attrition dataset and describe the analysed sample.